

120 Days of Quantitative Finance

Week 8: Volatility Models & Returns Forecasting

Day 50: Understanding Volatility in Finance

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1. Definition of Volatility

Volatility measures the dispersion of returns:

$$\sigma = \sqrt{Var(r_t)}$$

It represents the uncertainty or risk associated with asset returns.

2. Discrete-Time Return Model

$$r_t = \mu + \epsilon_t$$

where:

$$\epsilon_t \sim N(0, \sigma^2)$$

In classical models, σ^2 is assumed constant.

3. Time-Varying Volatility

Empirical evidence shows volatility is not constant.

$$Var(r_t | \mathcal{F}_{t-1}) = \sigma_t^2$$

This phenomenon is known as conditional heteroskedasticity.

4. Stylized Facts of Volatility

4.1 Volatility Clustering

$$Corr(r_t^2, r_{t-k}^2) > 0$$

Large shocks tend to be followed by large shocks.

4.2 Heavy Tails

Return distributions exhibit excess kurtosis:

$$Kurtosis(r_t) > 3$$

4.3 Leverage Effect

Negative returns often increase future volatility more than positive returns.

5. Continuous-Time Volatility

Under Geometric Brownian Motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Volatility controls the diffusion intensity.

6. Unconditional vs Conditional Volatility

Unconditional volatility:

$$\sigma^2 = E(r_t^2)$$

Conditional volatility:

$$\sigma_t^2 = E(r_t^2 | \mathcal{F}_{t-1})$$

This distinction motivates ARCH and GARCH models.

7. Realized and Implied Volatility

Realized volatility is calculated from historical returns.

Implied volatility is extracted from option prices and reflects market expectations.

Conclusion

Volatility is central to risk management, option pricing, and financial modeling. Empirical evidence shows volatility is time-varying, clustered, and asymmetric, motivating the development of dynamic volatility models.